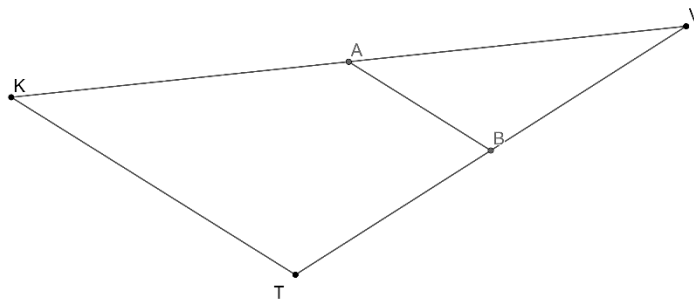


Geometry
Unit 11
Lesson 8 Homework

Name **Answer Key**
Date _____
Period _____

Use this information to answer questions 1 through 5.

Triangle KVT is shown, where $\overline{AB}$ is the midsegment. $KT = 2x + 4$, $AB = \frac{1}{2}x + 3$, $AV = \frac{5}{2}y - 6.5$, $KA = \frac{1}{2}y + 3.5$, $VT = 3z - 7$, and $TB = 2x + 2.1$.



1. Find the value of x .

$$\begin{aligned} 2\left(\frac{1}{2}x + 3\right) &= 2x + 4 \\ x + 6 &= 2x + 4 \\ x &= 2 \end{aligned}$$

2. Find the value of y .

$$\begin{aligned} \frac{1}{2}y + 3.5 &= \frac{5}{2}y - 6.5 \\ 3.5 &= 2y - 6.5 \\ 2y &= 10 \\ y &= 5 \end{aligned}$$

3. Find the value of z .

$$\begin{aligned} 2(2x + 2.1) &= 3z - 7 \\ 2(2(2) + 2.1) &= 3z - 7 \\ 2(6.1) &= 3z - 7 \\ 12.2 &= 3z - 7 \\ 3z &= 19.2 \\ z &= 6.4 \end{aligned}$$

4. Find the length of the midsegment, $\overline{AB}$.

$$AB = \frac{1}{2}(2) + 3 = 4$$

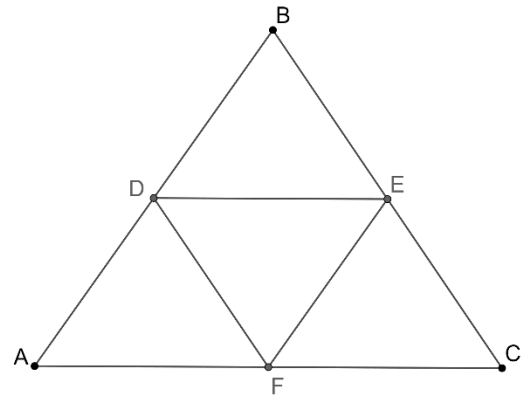
5. Find the length of $\overline{KV}$.

$$\begin{aligned} KV &= KA + AV \\ KV &= \frac{1}{2}(5) + 3.5 + \frac{5}{2}(5) - 6.5 = 12 \end{aligned}$$

Use this information to answer questions 6 through 10.

Triangle ABC is shown, where D , E , and F are midpoints.

$DE = 2x - 4$, $AC = 4x - 8$, $DF = x + 3$, $BC = 3x + 1$,
 $FE = 3x - 6$, and $AB = 5x - 7$.



6. Find the value of x .

$$\begin{array}{ll} 2(x + 3) = 3x + 1 & 2(3x - 6) = 5x - 7 \\ 2x + 6 = 3x + 1 & 6x - 12 = 5x - 7 \\ x = 5 & x = 5 \end{array}$$

7. Find the length of $\overline{DF}$.

$$DF = 5 + 3 = 8$$

8. Find the length of $\overline{EF}$.

$$EF = 3(5) - 6 = 9$$

9. Find the length of $\overline{DE}$.

$$DE = 2(5) - 4 = 6$$

10. Find the perimeter of triangle ABC .

$$\begin{array}{l} P = 2(EF) + 2(DF) + 2(DE) \\ P = 2(9) + 2(8) + 2(6) = 46 \end{array}$$